

Day 1: What is AWS? Why Cloud, Six Pillars, Careers and Your First Practical

Saturday, 10 October 2026 | 7:00-9:00 PM IST | Cloud Unlock x AWS Student Builder Group, MIT ADT University | Trainer: Gangadhar Ure

1. Goal of Today

By the end of this class you can explain what AWS is and why companies use it, open the AWS Console confidently, switch Regions, find your Free Tier and billing pages, and run your first AWS CLI command in CloudShell. No prior cloud knowledge is needed.

You will learn	You will do
What problem the cloud solves	Create or prepare a Free-plan AWS account
Six advantages of cloud computing	Tour the Console and switch Region
IaaS vs PaaS vs SaaS	Find Billing and Free Tier usage
Four ways to access AWS	Run your first CLI commands in CloudShell

2. Before Class Checklist

- A laptop with a modern browser (Chrome, Edge or Firefox).
- A personal email address you control (do NOT use a shared or college-lab email).
- A mobile phone for SMS/OTP verification and, later, an authenticator app.
- A payment card that works online. AWS needs it to verify your identity even on the Free plan. Never show it on screen or in screenshots.
- If you already have an account: sign in as your own user and check it opens.

Note: The Free plan gives USD 100 in credits on sign-up, plus up to USD 100 more by completing activities. It lasts 6 months or until credits run out. Terms change, so always check the Billing page and aws.amazon.com/free for today's rules.

3. Why Cloud? The Problem in One Story

Imagine your college fest website. For 360 days it gets 50 visitors a day. On registration day it gets 50,000. Without cloud you must buy servers for the peak, wait weeks to set them up, and pay for them even when idle. With cloud you rent what you need, in minutes, and pay only for what you use.

Traditional (own servers)	Cloud (AWS)
Big upfront cost (capex)	Pay as you go (opex)
Weeks to buy and set up	Minutes to launch
Guess capacity in advance	Scale up and down on demand
You maintain buildings, power, hardware	AWS maintains the data centres
Hard to serve users in other countries	Deploy close to users in many Regions

Five real problems the cloud solves

Problem	Without cloud	With AWS	Service you will meet
Scaling (autoscaling)	Buy servers for the peak; they sit idle the rest of the year	Capacity grows when traffic grows and shrinks when it falls	EC2 Auto Scaling, Elastic Load Balancing
Cost	Large upfront purchase, guessing capacity	Pay only for what you use; budgets and alerts	AWS Budgets, Cost Explorer, pricing models
Availability	One data centre failure = site down	Spread across Availability Zones and Regions	Multi-AZ, Route 53, CloudFront
Security	You secure buildings, hardware and software alone	Shared responsibility: AWS secures the cloud, you secure what you put in it	IAM, KMS, Security Groups, Shield
Reliability	Backups and recovery are manual and slow	Automated backups, replication, quick recovery	S3, RDS backups, AWS Backup

Story to remember: fest registration opens at 9 AM. Autoscaling adds servers at 9:00 and removes them at 11:00 (cost). Two Availability Zones keep the site up if one fails (availability). IAM and encryption protect student data (security). Automated backups bring data back if someone deletes it (reliability).

Six advantages of cloud computing (exam favourite)

#	Advantage	Plain meaning
1	Trade fixed expense for variable expense	No upfront purchase; pay for what you use
2	Benefit from massive economies of scale	AWS buys at huge volume, so prices come down
3	Stop guessing capacity	Scale to demand instead of guessing
4	Increase speed and agility	Resources in minutes, so experiments are cheap
5	Stop spending money running and maintaining data centres	Focus on your product, not hardware
6	Go global in minutes	Deploy in multiple Regions near your users

Cloud service models

Model	You manage	Example
IaaS	OS, apps, data (provider gives servers, storage, network)	Amazon EC2
PaaS	Your app and data (provider runs the platform)	AWS Elastic Beanstalk, Amazon RDS
SaaS	Just use it	Gmail, Microsoft 365, Salesforce

Deployment models

- Cloud: everything runs on the cloud provider.
- On-premises (private cloud): everything runs in your own data centre.
- Hybrid: a mix of both, connected together. Common when companies migrate step by step.

Exam shortcut: Pay-as-you-go + no upfront cost + scale quickly = cloud benefit. Keywords 'own data centre plus AWS together' = hybrid. 'You manage the OS' = IaaS (EC2).

3A. The Six Pillars of the AWS Well-Architected Framework

The Well-Architected Framework is AWS's checklist for building good systems in the cloud. It has six pillars. They are tested in the exam and used in real design reviews.

Pillar	Question it asks	Benefit to the business	Simple example
Operational Excellence	Can we run, monitor and improve it?	Fewer outages, faster fixes	Automate deployments, watch metrics
Security	Is data and access protected?	Customer trust, compliance	Least-privilege IAM, encryption
Reliability	Does it recover and keep working?	Less downtime	Multi-AZ, backups, tested recovery
Performance Efficiency	Are we using the right resources?	Fast for users, no waste	Right instance type, caching
Cost Optimization	Are we paying only for value?	Lower bills	Right-size, turn off idle, use Budgets
Sustainability	Are we minimising environmental impact?	Lower energy, lower cost	Efficient resources, scale down when idle

Exam shortcut: Pillar names: Operational Excellence, Security, Reliability, Performance Efficiency, Cost Optimization, Sustainability. Memory trick: O-S-R-P-C-S, 'Our Servers Rarely Pause, Cost Stays'. Question says 'recover from failure' = Reliability. 'Reduce spend' = Cost Optimization. 'Protect data' = Security.

3B. Certifications and the Job Market

AWS certifications prove your skills to employers. Cloud Practitioner is the first step. After it you choose a path based on the job you want.

Level	Certification	Good for roles such as
Foundational	Cloud Practitioner (CLF-C02), the series goal	Anyone starting in cloud, students, sales, support, managers
Foundational	AI Practitioner	AI/ML beginners
Associate	Solutions Architect Associate	Cloud engineer, architect track
Associate	Developer Associate	Cloud developer
Associate	SysOps / CloudOps Engineer Associate	Operations, cloud support
Associate	Data Engineer Associate	Data engineering
Professional / Specialty	Solutions Architect Professional, Security Specialty and others	Senior roles and specialisation

- Roles that use AWS skills: Cloud Support Associate, Cloud Engineer, DevOps Engineer, Solutions Architect, Data Engineer, Security Engineer.
- Employers look for hands-on skills plus a certification. That is why every class has a practical and you will build a portfolio on GitHub.
- Check aws.amazon.com/certification for the current list of exams, prices and renewal rules; they change over time.
- Cloud Practitioner exam: 65 questions, 90 minutes, pass mark 700 out of 1000. Domains: Cloud Concepts 24%, Security and Compliance 30%, Cloud Technology and Services 34%, Billing, Pricing and Support 12%.

3C. How to Create an AWS Account (step by step)

Guardrail: Use your own personal email. Never share OTP, password or card details. Use a unique password and keep your phone ready.

- 1 Open aws.amazon.com in your browser and choose Create an AWS Account.
- 2 Enter your root user email address and an AWS account name (for example cloudadhar-learning). Choose Verify email address.
- 3 Enter the verification code sent to your email.
- 4 Create a strong, unique root user password and confirm it.
- 5 Choose an account plan: the Free plan (credits, safe for learning) or the Paid plan. For this series choose Free, unless your account already exists.
- 6 Enter your contact information and choose Personal (not Business) for learning. Accept the customer agreement.
- 7 Add a payment method (card). AWS uses it to verify identity. Free plan credits cover learning usage. Never share the card details.
- 8 Verify your phone number with the SMS or voice code.
- 9 Choose the Basic support plan (free).
- 10 Wait for the account to activate (usually minutes, sometimes up to 24 hours). Sign in as Root user with your email.
- 11 Check the Billing page for your credits, then continue to the Console tour. Tomorrow we turn on MFA.

Note: The sign-up screens and plan names can change. If a screen looks different from this list, follow the on-screen text and ask in chat.

4. Four Ways to Access AWS

Method	What it is	Best for
Management Console	Web browser UI	Learning, exploring, one-off tasks
AWS CLI	Commands in a terminal	Quick tasks and scripts
SDKs / APIs	Code libraries (Python, Java, JavaScript...)	Applications calling AWS
Infrastructure as Code	CloudFormation, CDK, Terraform	Repeatable, version-controlled setups

5. Practical 1: Console Tour (about 18 min)

Guardrail: Never share your root password, OTP codes, card details, 12-digit account ID, or any access key in chat, screenshots or GitHub. Blur or crop before sharing evidence.

- 1 Sign in to the Console (Root user for today only; we create a safer user from Day 3). Top-right shows your Region name. Note it.
- 2 In the search bar, type 'EC2' and open the service. Go back to Console Home and type 'S3'. Notice services are grouped by category.
- 3 Click the Region selector. Switch to Asia Pacific (Mumbai) ap-south-1, then US East (N. Virginia) us-east-1, then back to Mumbai.
- 4 Open Billing and Cost Management from the account menu. Find Free Tier (or Credits) and note remaining credits.

- 5 Open the Services menu. Name one service each in Compute, Storage, Database, Networking and Security.

Expected result: You can sign in, you can name the Region you are in, and you can find your credits/Free Tier page.

6. Practical 2: First CLI Commands in CloudShell (about 12 min)

AWS CloudShell is a free browser-based terminal with the AWS CLI already installed and already signed in with your Console credentials. No installation, no access keys.

- 1 In the Console, make sure the Region is Asia Pacific (Mumbai). Click the CloudShell icon in the top bar (terminal symbol).
- 2 Wait for the prompt to appear. Run the commands below one by one.

```
aws --version
aws configure list
aws ec2 describe-regions --query "Regions[].RegionName" --output table
aws ec2 describe-availability-zones --query "AvailabilityZones[].ZoneName" --output text
```

Command	What it tells you
aws --version	CLI is installed and which version
aws configure list	Where credentials and Region come from (CloudShell sets them for you)
aws ec2 describe-regions	Regions enabled for your account
aws ec2 describe-availability-zones	Availability Zones in the Region you are in

Privacy tip: The command 'aws sts get-caller-identity' prints your 12-digit account ID. Run it if you like, but never paste the output publicly.

Expected result: You see a CLI version, a list of Region names including ap-south-1, and 3 or more AZ names such as ap-south-1a (check the live count in your own output).

7. Which Service Where? (Preview)

Need	AWS service (we cover each later)
Run a server	Amazon EC2
Store files	Amazon S3
Relational database	Amazon RDS / Aurora
Run code without servers	AWS Lambda
Control who can do what	AWS IAM
Watch cost	AWS Budgets / Cost Explorer

8. Evidence Checklist (post in the group)

- Screenshot of the Console Home with your Region visible (account ID blurred).
- Screenshot of the CLI output of aws ec2 describe-regions (no account ID).

- One sentence: 'AWS is useful for ____ because ____'.

9. Quick Quiz (7 questions, answers below)

- 1 Which is a benefit of cloud computing? A) Large upfront hardware purchase B) Pay-as-you-go pricing C) Buying capacity for peak D) Managing data centre power
- 2 In which model do you manage the operating system? A) SaaS B) PaaS C) IaaS D) None
- 3 Which tool is a browser-based terminal with the AWS CLI preinstalled? A) CloudFormation B) CloudShell C) CloudTrail D) CloudFront
- 4 Which is NOT a way to access AWS? A) Console B) CLI C) SDK D) Pen drive
- 5 Company runs some apps in its own data centre and some on AWS. This is: A) Cloud-native B) Hybrid C) Serverless D) Edge
- 6 Which Well-Architected pillar is about recovering from failure? A) Cost Optimization B) Reliability C) Sustainability D) Performance Efficiency
- 7 How many pillars does the Well-Architected Framework have? A) 4 B) 5 C) 6 D) 7

Answers: 1-B, 2-C, 3-B, 4-D, 5-B, 6-B, 7-C

10. Homework (30 min)

- Read the 'Six advantages of cloud computing' table twice and explain it in your own words to a friend.
- Open Console Home > Services and list 10 services you have never heard of. We will meet many of them.
- Come ready for Day 2: Regions, Availability Zones, and securing your account with MFA.

11. Cleanup

Day 1 creates no billable resources. Nothing to delete. Close CloudShell when finished.

12. References

- AWS Certified Cloud Practitioner (CLF-C02) Exam Guide, Domain 1: Cloud Concepts.
- AWS Free Tier: aws.amazon.com/free
- AWS CloudShell documentation: docs.aws.amazon.com/cloudshell